From fitted Hawkes parameters to interpretable quantities: noise, missing calls and the drivers of the call rate

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Abstract

We fit a log-Gaussian Cox background with self-exciting (Hawkes) triggering to North Atlantic right whale upcall detections at three passive acoustic buoys in southern New England. This note describes three derived quantities we propose to report, each a functional of the posterior. They are the multiplicative effect of a fixed increase in ambient noise on the detected call rate; a counterfactual count of calls "missing" at noisy times; and a day-by-day multiplicative decomposition of the background rate. We give the definitions, the computations, the current estimates and the assumptions each one leans on. We also set out three extensions we would like feedback on: a sonar-equation benchmark for the noise slope, a noise-dependent branching ratio as a model-based handle on masking versus behavior, and an honest attribution of the seasonal trend.

Status of the fits. COX01 is the clean-data refit (100k iterations, burn-in 75k). NS01 is the chain extended to 150k with burn-in 100k; its $-2\log L$ trace is flat over the retained draws. NS02 is still the 100k chain with burn-in 50k. That chain had not converged, and a 150k extension is running, so the NS02 numbers below are provisional. Every functional is computed from 2,000 equally spaced post-burn-in draws.

1 Model

Let $t_1 < \cdots < t_n$ be detected upcall times (minutes) on $[0, T]$, with T covering 1 October 2021 to 1 March 2022 (151 days). We observe $n = 6{,}098$ calls at COX01, 13,017 at NS01 and 33,611 at NS02. The conditional intensity is

$$\lambda(t \mid \mathcal{H}_t) = \mu(t) + \sum_{t_i < t} \alpha\, e^{-\eta(t - t_i)}, \qquad \mu(t) = \exp\{\boldsymbol{x}(t)^\top \boldsymbol{\beta} + \delta\, W(t)\}. \tag{1}$$

The kernel is unnormalized, so the branching ratio (the expected number of calls directly triggered by one call) is $m = \alpha/\eta$. We need $m < 1$. The expected cluster size is $1/(1 - m)$, and the half-life of the excitation is $\log 2/\eta$.

Background. $\boldsymbol{x}(t)$ contains an intercept; ambient noise $x_n(t)$ (Section 2); daily sea-surface temperature $x_s(t)$ (NOAA OISST, standardized); and sine and cosine pairs at periods of 1, 7, 14, 30 and 60 days. That makes $p = 13$ columns. W is a zero-mean, unit-variance Gaussian process with exponential covariance $\mathrm{Cov}\{W(s), W(s')\} = \exp(-|s - s'|/\rho)$, an Ornstein–Uhlenbeck process. The range is fixed at $\rho = 60$ min, so the correlation is e^{-3} at 3 h. $\delta > 0$ scales it. The background is

evaluated at knots $s_0, \ldots, s_M$ spaced 20 min apart ($M \approx 10{,}800$) and interpolated linearly between them. The compensator $\int_0^T \mu$ is computed by the trapezoid rule on the same knots.

Priors and computation. $\boldsymbol{\beta} \sim N(\mathbf{0}, 100\, I)$, $\log \delta \sim N(0, 100)$, $\alpha \sim \text{Gamma}(0.001, 0.001)$ and $\eta \sim U(0.15,\, 3/\min_i(t_{i+1} - t_i))$ per minute. The sampler augments the data with the latent branching structure (each call is either an immigrant or the offspring of a specific earlier call), drawn by Gibbs. Given the branching structure, α is conjugate, with shape $0.001 + n_{\text{off}}$ and rate $0.001 + \eta^{-1} \sum_i (1 - e^{-\eta(T - t_i)})$. $\boldsymbol{\beta}$, $\log \delta$ and η are updated by adaptive random-walk Metropolis (target acceptance 0.234), and W at the knots by elliptical slice sampling.

A posterior predictive check. Under (1), ignoring edge effects (the half-life is about 2 min, far shorter than any interval we sum over), the expected number of calls in a set A is

$$\mathbb{E}N(A) \approx \frac{1}{1 - m} \int_A \mu(t)\, dt. \tag{2}$$

Over the whole window the posterior median of (2) is 6,086, 13,014 and 33,589, against observed counts of 6,098, 13,017 and 33,611. That agreement is a property of a well-mixed fit, not evidence for the noise model, but it confirms the functionals below are on the right scale. Table 1 gives the parameter summaries.

Table 1: Posterior medians (95% equal-tailed intervals) of the parameters used below. β_n and β_s are coefficients on the standardized noise and SST covariates. NS02 is provisional.

	COX01	NS01	NS02
α	0.203 (0.185, 0.223)	0.223 (0.209, 0.238)	0.294 (0.283, 0.305)
η (min^{-1})	0.340 (0.308, 0.379)	0.386 (0.361, 0.419)	0.494 (0.453, 0.527)
$m = \alpha/\eta$	0.594 (0.561, 0.634)	0.576 (0.546, 0.604)	0.596 (0.567, 0.641)
$1/(1-m)$	2.47 (2.28, 2.73)	2.36 (2.20, 2.53)	2.47 (2.31, 2.79)
$\log 2/\eta$ (min)	2.04 (1.83, 2.25)	1.79 (1.66, 1.92)	1.40 (1.32, 1.53)
δ	2.03 (1.89, 2.15)	1.81 (1.73, 1.90)	1.60 (1.54, 1.66)
β_n	−0.590 (−0.705, −0.466)	−1.477 (−1.628, −1.333)	−0.796 (−0.889, −0.727)
β_s	−2.84 (−3.12, −2.54)	−1.76 (−1.97, −1.62)	−1.39 (−1.49, −1.31)

2 Putting the noise coefficient on the decibel scale

The noise covariate is the RMS noise level from the recorder, rescaled per deployment and then standardized over the knots: $x_n(s) = \{y(s) - \bar{y}\}/\text{sd}(y)$. The RMS level in dB re 1 μPa is an affine function of y; a regression on the raw one-minute files recovers it with $R^2 > 0.999$ at every buoy. So

$$x_n(s) = \frac{L(s) - \bar{L}}{s_L},$$

where L is the noise level in dB and s_L is its standard deviation over the knots: 4.00 dB (COX01), 6.02 dB (NS01) and 5.30 dB (NS02). The coefficient β_n/s_L is therefore the change in $\log \mu$ per dB of noise. The RMS level is measured in the right whale upcall band, at the hydrophone.

3 Fig. 4: the effect of a fixed increase in noise

Estimand. For an increase of Δ dB at a fixed time, holding everything else fixed,

$$R(\Delta) = \frac{\mu(t;\, L+\Delta)}{\mu(t;\, L)} = \exp\!\left(\frac{\beta_n \Delta}{s_L}\right), \tag{3}$$

which doesn't depend on t. By (2), $R(\Delta)$ is also the ratio of expected *total* detected calls, triggered calls included, as long as noise acts only on μ. Section 4.1 questions that assumption. We take $\Delta = 5\,\text{dB}$, which is about one s_L at each buoy, so the comparison is between sites on the same physical scale rather than the same standardized one. The posterior of R comes from the draws of β_n, with s_L treated as known.

Table 2: Noise effect on the detected call rate. The slope is β_n/s_L, in log-rate per dB. $R(\Delta)$ is the multiplicative effect of $+\Delta$ dB.

	s_L (dB)	slope (per dB)	$R(1\,\text{dB})$	$R(5\,\text{dB})$
COX01	4.00	−0.148 (−0.176, −0.117)	0.863 (0.839, 0.890)	0.478 (0.415, 0.558)
NS01	6.02	−0.245 (−0.271, −0.221)	0.782 (0.763, 0.801)	0.293 (0.259, 0.331)
NS02	5.30	−0.150 (−0.168, −0.137)	0.861 (0.846, 0.872)	0.472 (0.433, 0.504)

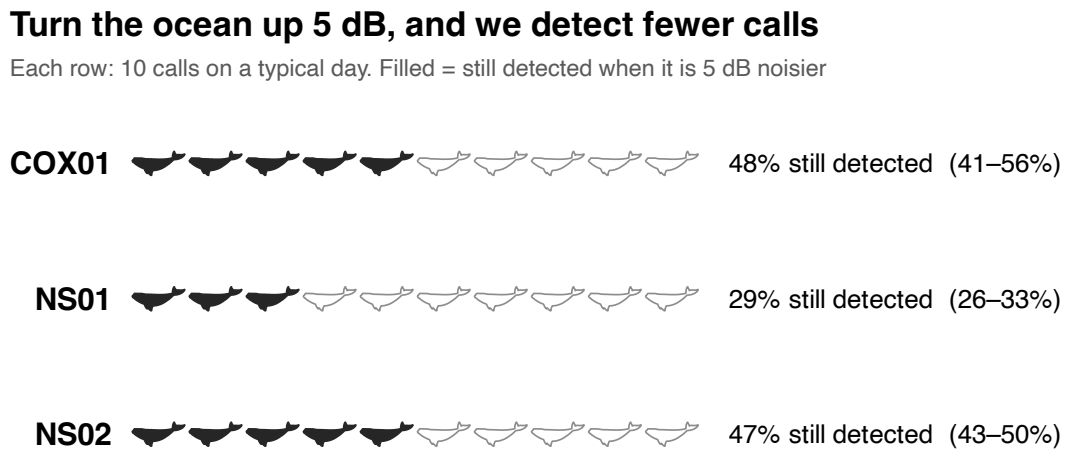


Figure 1: Fig. 4 (talk version). Each row shows $10\,R(5\,\text{dB})$ rounded to whole calls. The text gives the posterior median and 95% interval of $R(5\,\text{dB})$.

Proposal: a sonar-equation benchmark for the slope. A per-dB slope can be compared with what pure masking predicts. In the passive sonar equation, a call is detected when $\text{SL} - \text{TL}(r) - L \geq \text{DT}$, with transmission loss $\text{TL}(r) = k \log_{10} r$. The maximum detection range is then $r_{\max} = 10^{(\text{SL}-L-\text{DT})/k}$. If the detected call rate is proportional to the area within $r_{\max}$,

$$\frac{\partial \log \mu}{\partial L} = -\frac{2 \ln 10}{k} = \begin{cases} -0.230 & k = 20 \text{ (spherical)}, \\ -0.307 & k = 15 \text{ (“practical”)}, \\ -0.461 & k = 10 \text{ (cylindrical)}. \end{cases} \tag{4}$$

Assumptions behind (4).

A1. *Detection depends only on the signal-to-noise ratio at the hydrophone.* Masking happens at the receiver, and L is measured at the receiver in the upcall band, so the covariate is the right quantity. This is the strongest assumption in the argument.

A2. *Transmission loss is a power law,* $k \log_{10} r$, the same in all directions, with negligible absorption at upcall frequencies. Shallow-water loss usually lies between cylindrical and spherical, and k varies with site and season.

A3. *Source level does not depend on noise.* SL and DT may vary from call to call with any distribution, provided it is independent of L: every call's $r_{\max}$ then scales by the same factor $10^{-\Delta/k}$.

A4. *Callers are spread uniformly in the plane, out beyond* $r_{\max}$.

A5. *The calling rate per animal does not depend on noise.*

The shape of the detection function does not matter. Let detection be probabilistic, with probability $g(\mathrm{SNR})$ for any function g. Under A2–A4 the detected area is

$$\int_0^\infty g(\mathrm{SL} - \mathrm{DT} - L - k \log_{10} r)\, 2\pi r\, dr = 10^{2(\mathrm{SL}-\mathrm{DT}-L)/k} \int_0^\infty g(-k \log_{10} \rho)\, 2\pi\rho\, d\rho,$$

by the substitution $r = 10^{(\mathrm{SL}-\mathrm{DT}-L)/k}\rho$. The remaining integral doesn't involve L, so the slope is exactly $-2 \ln 10/k$ whatever the detection curve. A smooth detection curve only matters in combination with a failure of A2 or A4.

Relaxing the assumptions. Three departures each change the slope in a known way.

- *Lombard compensation* (fails A3): if callers raise their source level by c dB per dB of noise, $r_{\max} \propto 10^{-(1-c)L/k}$. Tagged right whales show $c \approx 1$ at moderate noise (Parks et al. 2011).
- *A density gradient* (fails A4): if density falls with distance as r^{-a}, $0 \le a < 2$, the detected rate scales as $r_{\max}^{2-a}$. If $r_{\max}$ exceeds the area callers occupy, the slope tends to 0 at low noise (saturation).
- *Calling less in noise* (fails A5): if each animal's calling rate is proportional to e^{bL}, b adds directly to the slope.

Together,

$$\frac{\partial \log \mu}{\partial L} \approx -\frac{(1-c)(2-a)\ln 10}{k} + b, \tag{5}$$

one equation in four unknowns (k, a, c, b).

What the slopes say.

- NS01's slope, -0.245 (-0.271, -0.221), contains the spherical benchmark. It is consistent with pure masking at $k = 20$ ($c = a = b = 0$). It is equally consistent with some compensation offset by some reduction in calling.
- COX01 and NS02 are shallower (≈ -0.15). Under (5) there are three readings:
  - partial compensation, $c \approx 0.35$ at $k = 20$ or $c \approx 0.5$ at $k = 15$;
  - callers concentrated near the buoy, $a > 0$;

  - steeper effective transmission loss: pure masking with $k \approx 31$ gives -0.15.

- A slope *steeper* than any plausible masking slope would require $b < 0$: fewer calls made, not just fewer heard. No site shows that for $k \leq 20$.

From a frame to a test. We don't offer (4) as a test yet. Three additions would make it one:

- site-specific k from propagation modeling;
- a low-dimensional spline in L in place of the linear term, to detect saturation (curvature at low noise) directly;
- some handle on the density gradient a, for example from array localizations.

With k fixed, (5) bounds the combination of c and b. Two effects attenuate the fitted slope toward 0 and should be kept in mind. The covariate is a 1-minute RMS averaged to 20-minute knots, while masking acts call by call, so short noise spikes behave like measurement error. Orphaned answers (Section 4.1) inflate the apparent background most at noisy times.

4 Fig. 6: calls missing at noisy times

Estimand. Let x_q be the q-quantile of x_n over the knots of that buoy ($q = 0.1$). The counterfactual background caps the noise at x_q and leaves quieter times unchanged:

$$\mu_q(s) = \mu(s) \exp\big[\beta_n \{\min(x_n(s), x_q) - x_n(s)\}\big] \;\geq\; \mu(s). \tag{6}$$

For day d let $\Lambda_d = \int_d \mu$ and $\Lambda_d^q = \int_d \mu_q$ (trapezoid rule on the knots). Using (2), the expected detected calls are $\Lambda_d/(1-m)$ as fitted and $\Lambda_d^q/(1-m)$ in the quiet counterfactual. The gap splits into missing immigrant ("contact") calls, $\Lambda_d^q - \Lambda_d$, and the calls those would have triggered, $(\Lambda_d^q - \Lambda_d)\, m/(1-m)$. Every quantity is computed per posterior draw, jointly in $(\boldsymbol{\beta}, \delta, W, \alpha, \eta)$, and then summarized. These are posteriors of *expected* counts, not predictive distributions, so no Poisson or cluster variability is added.

Table 3: Season totals, posterior medians (95% intervals). "Missing" is the quiet total minus the fitted total. The share is the ratio of the two posterior medians. Noise levels are on the knots.

	$L_{0.1}$ / median L (dB)	fitted	quiet ($q = 0.1$)	missing	share
COX01	72.9 / 77.6	6,086 (5,737, 6,466)	11,456 (9,830, 13,119)	5,357 (3,856, 6,933)	47%
NS01	77.6 / 84.2	13,014 (12,477, 13,541)	30,738 (27,494, 34,533)	17,733 (14,698, 21,304)	58%
NS02	76.8 / 82.5	33,589 (32,749, 34,552)	87,399 (79,577, 98,646)	53,667 (46,244, 64,916)	61%

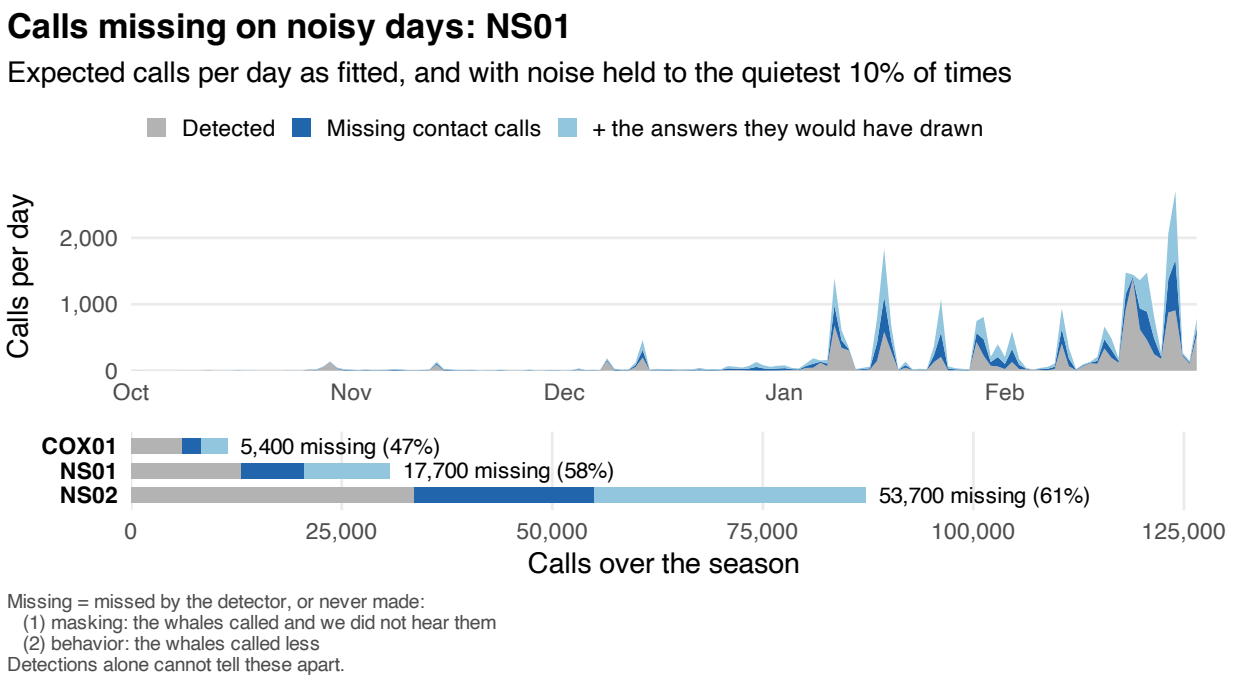


Figure 2: Fig. 6(i). NS01 daily posterior-median expected calls, fitted (gray) and under the capped-noise counterfactual (6). The dark band is missing immigrant calls; the light band is the calls they would have triggered. Lower panel: season totals for all three buoys.

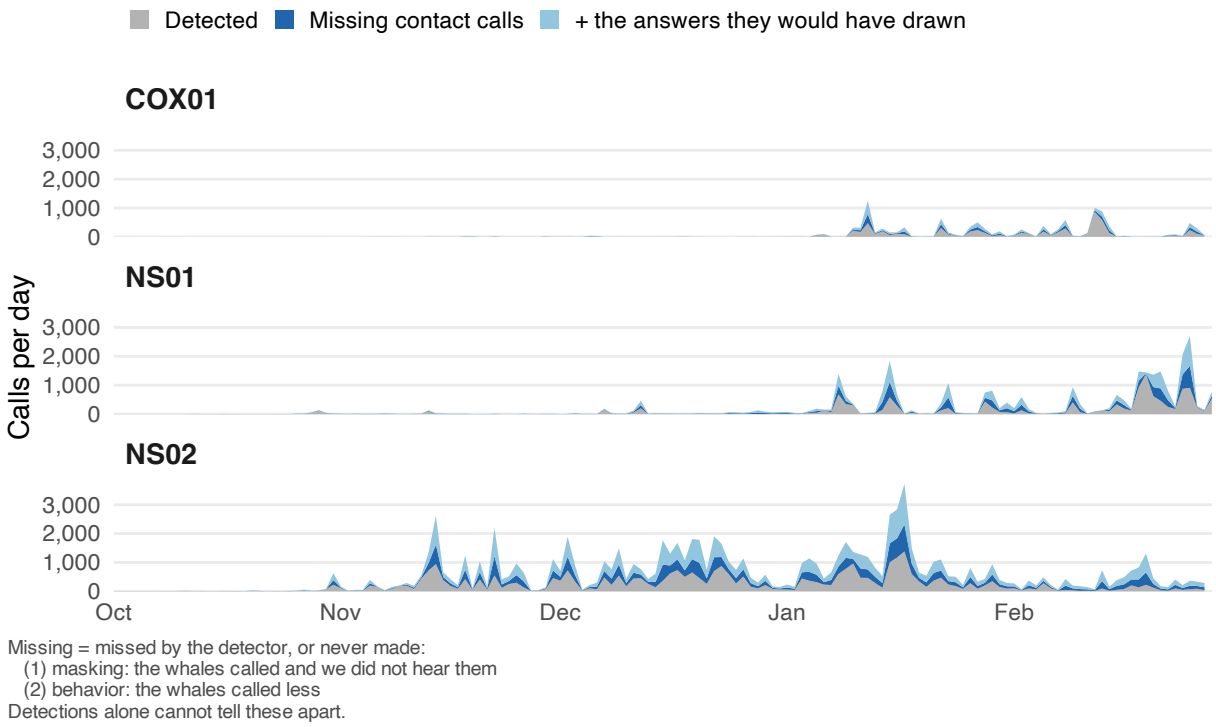


Figure 3: Fig. 6(ii). The same for the three buoys on a shared axis.

What the counterfactual does and doesn't assume.

- There's no extrapolation in x_n: x_q is inside the observed range, and the cap lowers the noise at 90% of the knots.
- It does lean on the log-linear form across the whole range. If the effect saturates at low noise (Section 3, A4), the linear model overstates Λ^q.
- Everything else, W included, is held at its fitted value. If W has absorbed residual structure correlated with noise, the attribution to noise is affected in a direction we can't sign a priori.
- The gap is "missed by the detector or never made". Detections alone don't separate masking from a behavioral reduction in calling. Section 4.1 suggests one model-based handle.

Proposed sensitivity analyses. Report the missing share as a function of $q \in \{0.1, 0.25, 0.5\}$, using the export's existing `PARAM_QUIET_Q` switch. Repeat the calculation under a spline-in-L noise effect. As an alternative estimand that doesn't depend on choosing a reference level, report the average marginal loss $-\int \partial \mathbb{E}N/\partial L \, dt$ in calls per dB.

4.1 Thinning, and a handle on masking versus behavior

If masking is the mechanism, it removes *every* call with some probability $p(t)$, triggered calls included. Let the true calls follow a Hawkes process with background $\mu^*(t)$ and branching ratio $m^* = \alpha^*/\eta$. The detector then sees a thinned Hawkes process.

Assumptions.

B1. *Detection is independent across calls, given the noise.* This is the critical assumption. If an answer comes from a different whale at an unrelated range, it is detected with probability $p(t)$ independently of its parent. If it is the same whale repeating in a bout, or a nearby whale, then a detected parent makes a detected answer very likely: same source level, same range, same noise. The model can't distinguish repeats from replies, so we don't know which case dominates.

B2. *p is roughly constant within a cluster.* The half-life is about 2 min, and the noise changes more slowly than that.

B3. *The thinned process is approximately Hawkes.* It isn't exactly: a detected answer whose parent was missed enters as an apparent immigrant (an "orphan").

Consequences under B1.

(a) A detected call's detected offspring number $p\,m^*$ on average, so the detected branching ratio falls with noise: $m(t) \approx p(t)\,m^*$. The current model holds m fixed, so under masking it is misspecified.

(b) Relative to the true immigrants, orphans number about $(1-p)\,m^*$. They inflate the apparent background most at noisy times, which pulls the fitted β_n toward 0.

(c) Both effects make the missing counts in Table 3 underestimates. A larger m at quiet times raises the quiet multiplier $1/(1-m_q)$, and the true background slope is steeper than fitted. Saturation (A4) pushes the other way, so the net direction is not guaranteed.

Proposed model. Let the noise at the parent's time scale its offspring: $\alpha_i = \alpha \exp\{\gamma\, x_n(t_i)\}$. Given the branching structure, call i's offspring count is Poisson with mean $\alpha e^{\gamma x_i}(1 - e^{-\eta(T-t_i)})/\eta$. So α stays conjugate, with full conditional

$$\alpha \mid \cdot \sim \text{Gamma}\Big(0.001 + n_{\text{off}},\ 0.001 + \eta^{-1} \textstyle\sum_i e^{\gamma x_i}\big(1 - e^{-\eta(T-t_i)}\big)\Big),$$

and the only new piece is a Metropolis step for γ. If the noise at the child's time matters more, a variant puts $e^{\gamma x(t)}$ inside the kernel integral. That is nearly identical here, because x_n barely changes within a 2-minute cluster.

Interpreting γ. If masking is the mechanism:

- $\gamma \approx \beta_n$: answers are thinned independently of their parents (different whales at unrelated ranges);
- $\gamma \approx 0$: answers are detected together with their parents (same-whale bouts);
- $\gamma/\beta_n \in [0, 1]$ measures roughly the fraction of answers whose detection is independent of their parent's.

If noise acts on behavior instead, $\gamma < 0$ would mean whales answer less in noise. So γ constrains the mechanism; it doesn't prove masking. Few calls are detected at high noise, so the posterior for γ may be wide exactly where it matters; its width will tell us.

5 Fig. 7: what moved the call rate

Decomposition. At each knot,

$$\log \mu(s) = \beta_0 + \underbrace{\beta_n x_n(s)}_{c_n(s)} + \underbrace{\beta_s x_s(s)}_{c_s(s)} + \underbrace{\textstyle\sum_j \beta_j h_j(s)}_{c_h(s)} + \underbrace{\delta W(s)}_{c_W(s)}.$$

For component c and day d let $\bar{c}(d)$ be the mean over the knots in d. Fig. 7 plots

$$F_c(d) = \exp\Big\{\bar{c}(d) - \tfrac{1}{D}\textstyle\sum_{d'} \bar{c}(d')\Big\},$$

the factor by which component c multiplied the background on day d, relative to its own season average. The product over components of $F_c(d)$ times a constant is the geometric-mean background on day d. The components use per-knot posterior medians, a plug-in summary; the median of the sum is not the sum of the medians, so the panels don't multiply exactly to the posterior-median background.

Answering. The bottom panel is the data-conditional ratio

$$A(d) = \frac{\sum_{s\in d}\{\mu(s) + \phi(s)\}}{\sum_{s\in d}\mu(s)}, \qquad \phi(s) = \sum_{t_i<s} \alpha e^{-\eta(s-t_i)},$$

on a 5-min grid, using the observed calls and plug-in posterior medians. It is left uncentered, so $A = 1$ on call-free days. Its geometric mean over days is 1.48 (COX01), 1.57 (NS01) and 2.11 (NS02). For NS01, μ, α and η in this panel still come from the 100k fit; that will be updated.

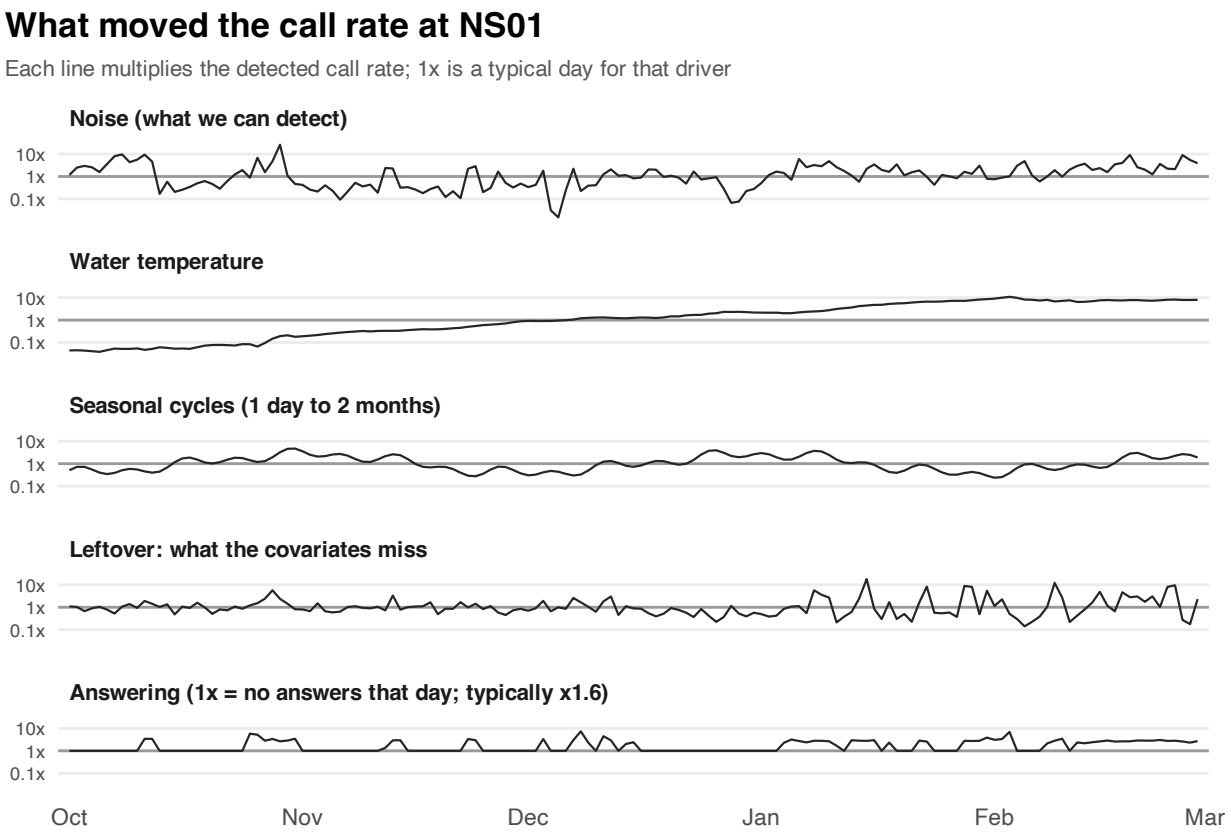


Figure 4: Fig. 7, NS01. Daily multiplicative factors $F_c(d)$ for the four background components, and the answering ratio $A(d)$. Log scale. COX01 and NS02 are in the Appendix.

Variance attribution. To summarize the panels we split the day-to-day variance of the daily-mean log background, $S(d) = \sum_c \bar{c}(d)$, into $\mathrm{Cov}\{\bar{c}(d), S(d)\}/\mathrm{Var}\{S(d)\}$. These shares sum to one and can be negative when components are negatively correlated (Table 4).

Table 4: Covariance shares of the variance of the log background, at daily means and at the 20-min knots. Plug-in posterior medians.

	daily means				knots			
	noise	SST	harmonics	GP	noise	SST	harmonics	GP
COX01 ($\mathrm{Var}\, S = 9.84$)	−0.02	0.79	0.18	0.06	0.00	0.62	0.15	0.22
NS01 ($\mathrm{Var}\, S = 6.90$)	0.29	0.50	0.08	0.14	0.29	0.37	0.06	0.29
NS02 ($\mathrm{Var}\, S = 2.71$)	−0.07	0.49	0.14	0.44	0.02	0.28	0.10	0.60

What Table 4 shows:

- Noise is a major driver of day-to-day variation only at NS01. A component's variance is β^2 times the variance of its covariate. NS01 has the largest per-SD effect (−1.48) and the most variable daily noise: the SD of daily-mean L is 4.9 dB, against 2.0 dB at COX01 and 3.7 dB at NS02.
- At COX01 and NS02 the noise contribution is negatively correlated with the rest of the background (−0.33 and −0.48 at daily means; −0.15 and −0.30 at knots). That cancels most

of its own variance, so its share is near zero at both scales, not just day to day.

- At NS02, daily noise and SST contributions correlate at -0.74. Both coefficients are negative, so louder days fall later in the season, as winter storms build while whales arrive. Much of the variation that identifies β_n there is sub-seasonal, and the split between noise and season is weakly identified.
- The GP share at NS02 (0.44 daily, 0.60 at knots) says there is a lot of residual structure on scales the covariates don't reach. We read it as a goodness-of-fit diagnostic as much as a driver.

Identifiability of the seasonal trend. The harmonics have periods of at most 60 days over a 151-day window. They can represent recurring rhythms but approximate a monotone trend only poorly. The GP has a 3-hour memory and can't carry one either. SST is the only smooth, monotone term, and over the window it correlates with time at -0.99, -0.98 and -0.96. In practice c_s is therefore the seasonal trend, and β_s should not be read as an effect per °C. With one season per site, temperature and time of year are confounded. We propose either replacing SST with an explicit smooth trend in time, or reporting $c_s + c_h$ together as "season", and keeping SST for when multiple seasons or decorrelated sites are available.

Proposed improvements. Compute $F_c(d)$ and $A(d)$ per draw and show pointwise 95% bands. Report the shares in Table 4 with posterior intervals. Given the correlations between components, consider a Shapley-type attribution, which is order-invariant, as an alternative to covariance shares.

6 Questions for discussion

1. Is the sonar-equation benchmark (4) a useful frame for the noise slope? How should uncertainty in k and the density gradient a be carried into (5)?
2. Detection belongs in the model. Is a noise-dependent kernel (Section 4.1) adequate, or should we fit the thinning explicitly, with undetected calls as latent variables?
3. Is the capped-noise expected count the right estimand for "calls lost to noise"? Or is the average marginal effect in calls per dB cleaner, since it avoids choosing a reference level?
4. NS01's η drifted across the first 100k iterations, with a matching shift between the GP background and the excitation. That is the usual clustering-versus-contagion trade-off. Would a reparameterization in (m, η), or a prior on δ, reduce it?

A Fig. 7 for COX01 and NS02

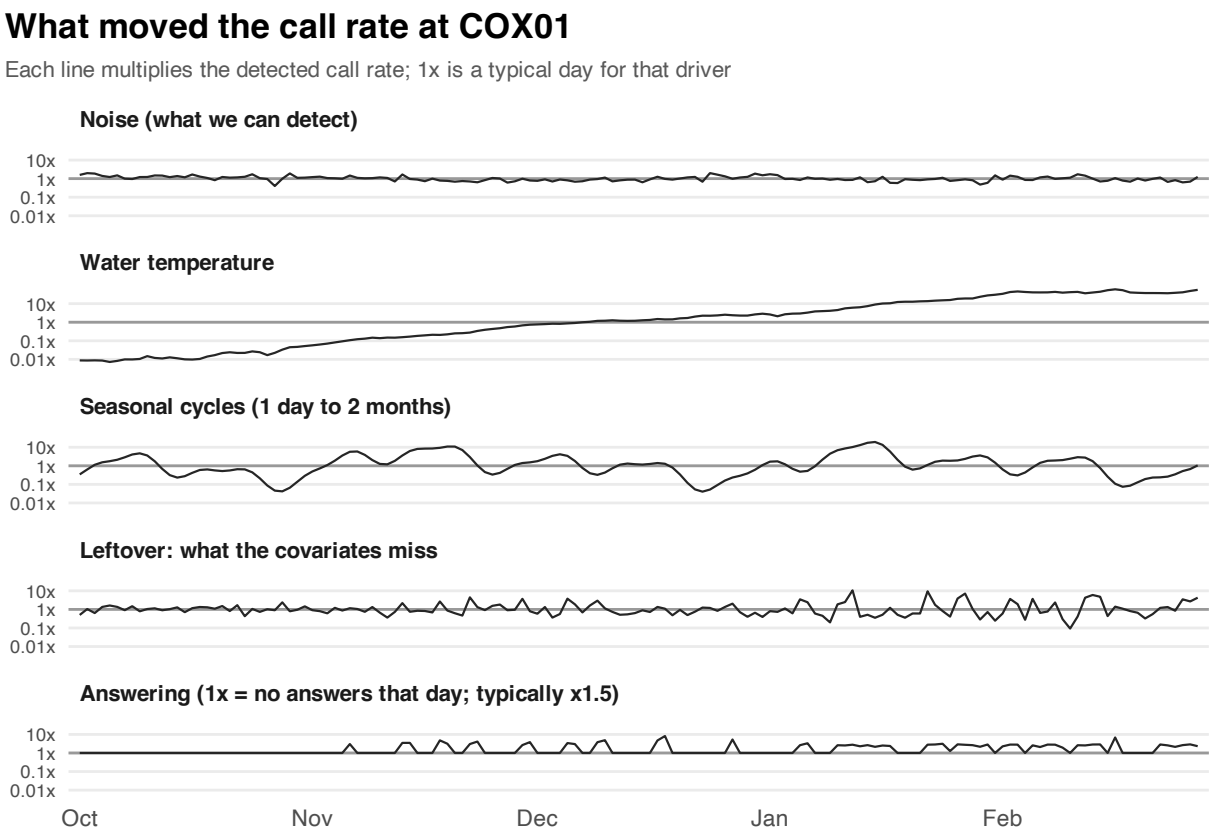


Figure 5: COX01.

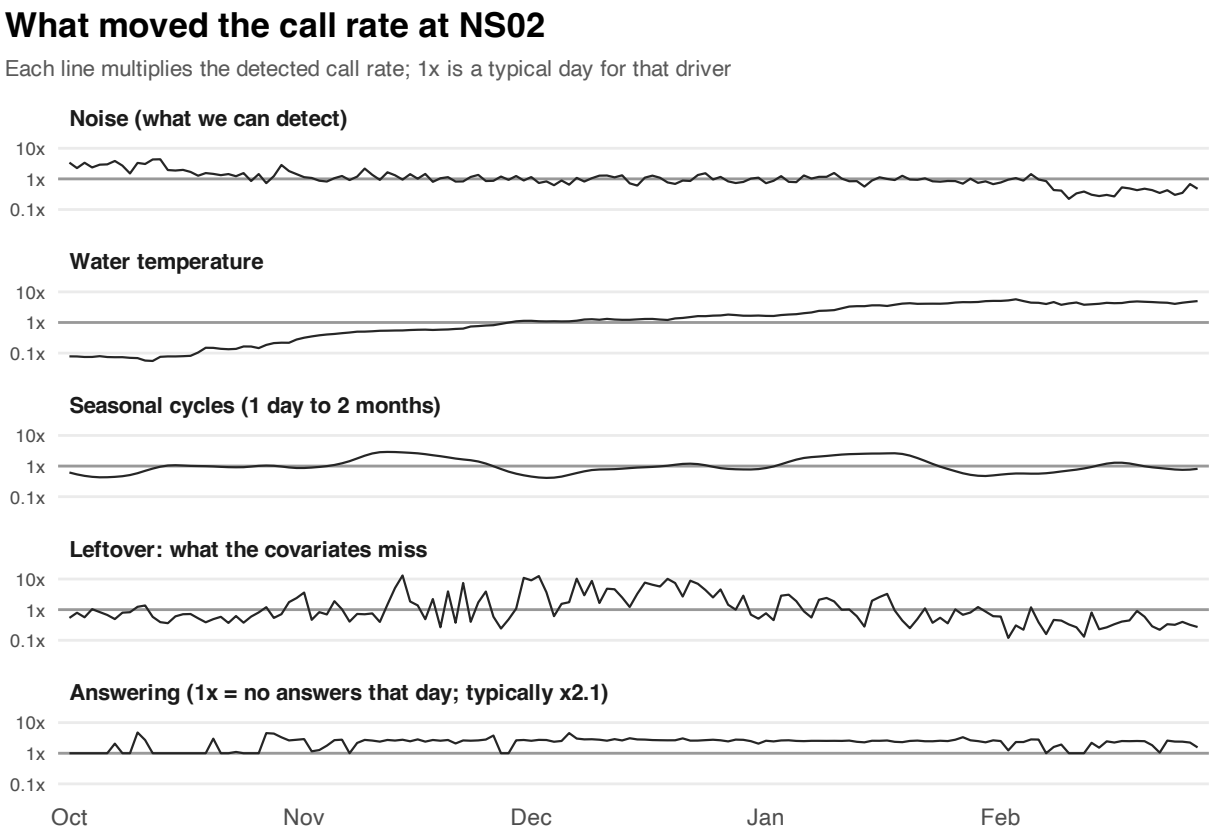


Figure 6: NS02 (provisional fit).

Reference. Parks, S. E., Johnson, M., Nowacek, D. and Tyack, P. L. (2011). Individual right whales call louder in increased environmental noise. *Biology Letters* 7(1), 33–35. doi:10.1098/rsbl.2010.0451.

Reproducibility. Cluster export: `10b_paramExport.R`. Figures: `14_volumeKnob.R`, `15_quietSeason.R` and `17_rateDrivers.R`, with the shared helpers in `src/talk_figs.R`.